2.3 Solving Trigonometric Equations (day 2)

Name _____ Period ____ Date _____

Pre-Calculus 2.3 (30-minute plan) · Student sheet

I can…

- use inverse trig functions to solve a trig equation and find all of its solutions.

Part 1 · Notes: fill in the blanks

arcsin gives an angle in _____; arccos gives an angle in _____; arctan gives an angle in _____.

sin x = k: x and _____ cos x = k: x and _____ tan x = k: x and _____

A negative calculator answer? Add _____ (or _____ for tangent) to land in [0, 2π). Calculator in _____ mode!

Part 2 · Examples (work them with the class)

Example 1

Solve on [0, 2π): $\sin x = 0.3$

Example 2

Solve on [0, 2π): $\cos x = -0.4$

Example 3

Solve on [0, 2π): $3\sin^2 x - 5\sin x - 2 = 0$

Part 3 · Partner practice

Partner practice

Solve on [0, 2π): $2\cos x = 1.5$

Part 4 · On your own: choose 2 of 3

Problem 1

Solve on $[0, 2\pi)$ (round to 4 decimals): $\sin x = 0.7$

Problem 2

Solve on $[0, 2\pi)$ (round to 4 decimals): $\tan x = -2$

Problem 3

Solve on $[0, 2\pi)$ (round to 4 decimals): $2\sin^2 x + 5\sin x - 3 = 0$